

7.16 The first derivative of the log marginal likelihood (7.97) is given by (7.100):

$$\frac{d\lambda(\alpha_i)}{d\alpha_i} = \frac{\alpha_i^{-1}s_i^2 - (q_i^2 - s_i)}{2(\alpha_i + s_i)^2}$$

Taking the second derivative:

$$\begin{aligned} \frac{d^2\lambda(\alpha_i)}{d\alpha_i^2} &= (\alpha_i^{-1}s_i^2 - (q_i^2 - s_i)) \cdot \frac{d}{d\alpha_i} \frac{1}{2(\alpha_i + s_i)^2} \\ &\quad + \frac{1}{2(\alpha_i + s_i)^2} \cdot \frac{d}{d\alpha_i} (\alpha_i^{-1}s_i^2 - (q_i^2 - s_i)) \\ &= (\alpha_i^{-1}s_i^2 - (q_i^2 - s_i)) \cdot \left(\frac{1}{2} \cdot \frac{(-2)}{(\alpha_i + s_i)^3} \right) \\ &\quad + \frac{1}{2(\alpha_i + s_i)^2} \cdot ((-1)\alpha_i^{-2}s_i^2) \\ &= -\frac{(\alpha_i^{-1}s_i^2 - (q_i^2 - s_i))}{(\alpha_i + s_i)^3} - \frac{(\alpha_i^{-2}s_i^2)}{2(\alpha_i + s_i)^2} \end{aligned}$$

Substituting the stationary point for α_i given by (7.101):

$$\begin{aligned} &= -\frac{\left(\left(\frac{s_i^2}{q_i^2 - s_i} \right)^{-1} s_i^2 - (q_i^2 - s_i) \right)}{\left(\left(\frac{s_i^2}{q_i^2 - s_i} \right) + s_i \right)^3} - \frac{\left(\left(\frac{s_i^2}{q_i^2 - s_i} \right)^{-2} s_i^2 \right)}{2 \left(\left(\frac{s_i^2}{q_i^2 - s_i} \right) + s_i \right)^2} \\ &= -\frac{((q_i^2 - s_i) - (q_i^2 - s_i))}{\left(\left(\frac{s_i^2 + q_i^2 s_i - s_i^2}{q_i^2 - s_i} \right) \right)^3} - \frac{((q_i^2 - s_i)^2)}{2 \left(\left(\frac{s_i^2}{q_i^2 - s_i} \right) + s_i \right)^2} \\ &= 0 - \frac{(q_i^2 - s_i)^2}{2 \left(\left(\frac{s_i^2}{q_i^2 - s_i} \right) + s_i \right)^2} \\ &= -\frac{1}{2} \left(\frac{(q_i^2 - s_i)}{\left(\left(\frac{s_i^2}{q_i^2 - s_i} \right) + s_i \right)} \right)^2 \end{aligned}$$

$$\begin{aligned}
&= -\frac{1}{2} \left(\frac{(q_i^2 - s_i)}{\left(\frac{s_i^2 + q_i^2 s_i - s_i^2}{(q_i^2 - s_i)} \right)} \right)^2 \\
&= -\frac{1}{2} \left(\frac{(q_i^2 - s_i)^2}{q_i^2 s_i} \right)^2
\end{aligned}$$

This result is always < 0 . The interior stationary point exists only when $q_i^2 > s_i$. If $q_i^2 = s_i$, (7.101) gives $\alpha_i \rightarrow \infty$, corresponding to the boundary of the parameter space rather than a finite stationary point.